

SSY236



Final Project: Perception, Reasoning, Control & Learning

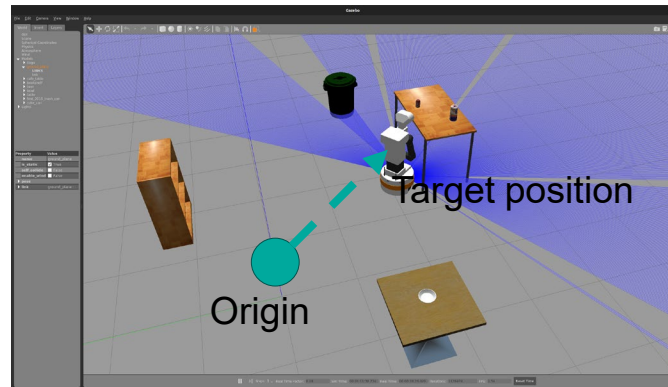
Karinne Ramirez-Amaro

Emmanuel Dean

Final Project

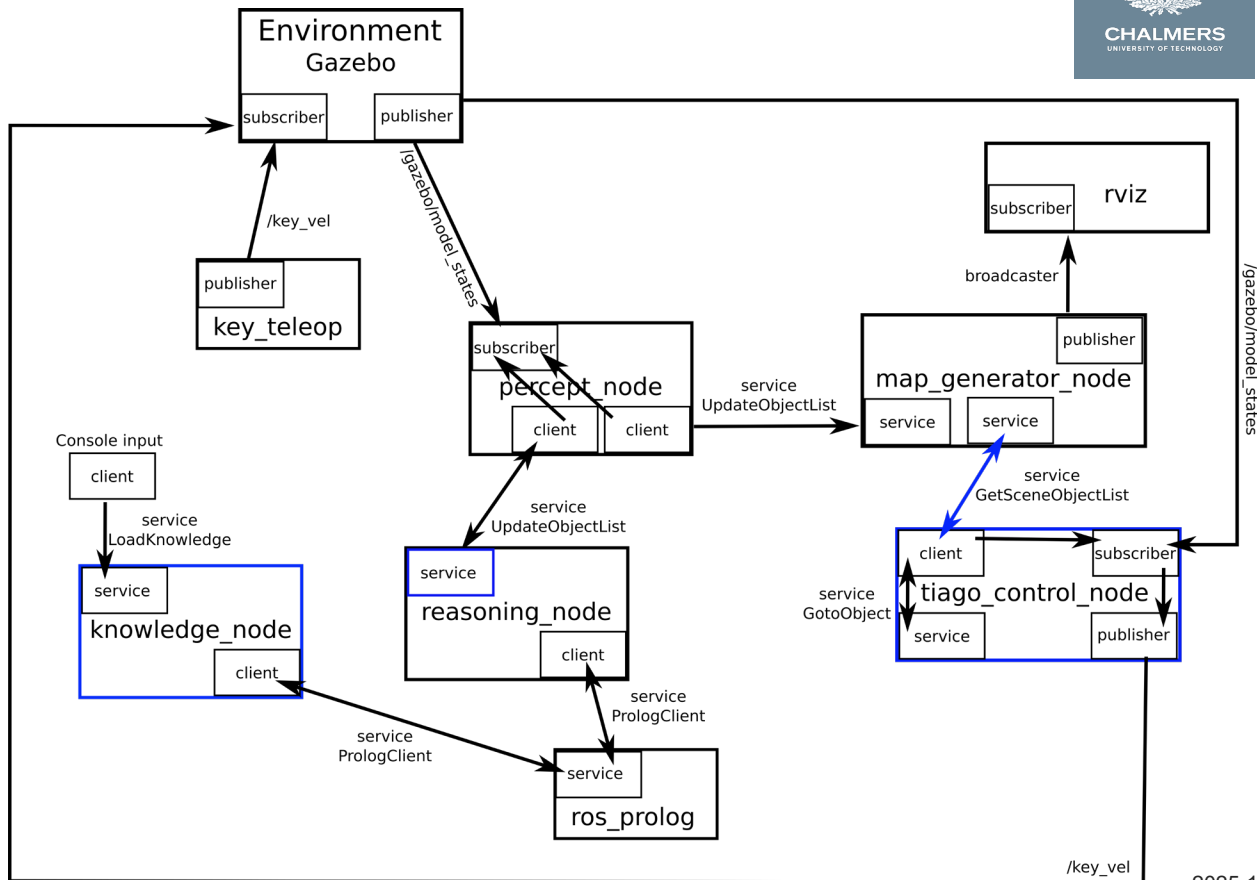
In the final project, you can accumulate a maximum of 32 points.

Goal: The robot should make autonomous decisions based on the learned knowledge. For example, the robot should learn that bottles are more likely located on top of tables. Then, based on this knowledge, the robot should make a decision to navigate to the big table to find out if the red bottle is there.



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As a starting point,
you can use your
solutions from
Assignment 04



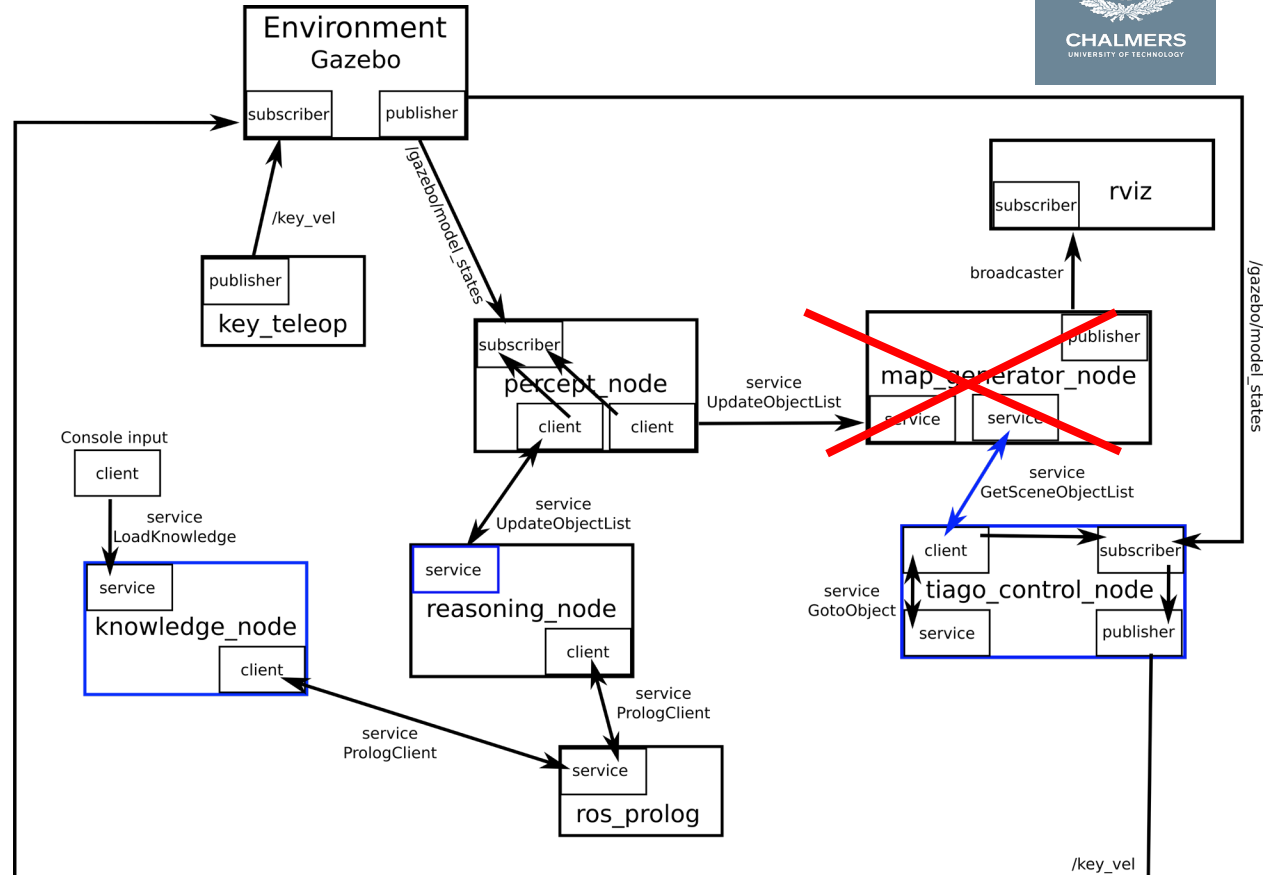
Final Project

1) For your final project, you should not use the “map_generator_node”.

This node should be removed entirely.

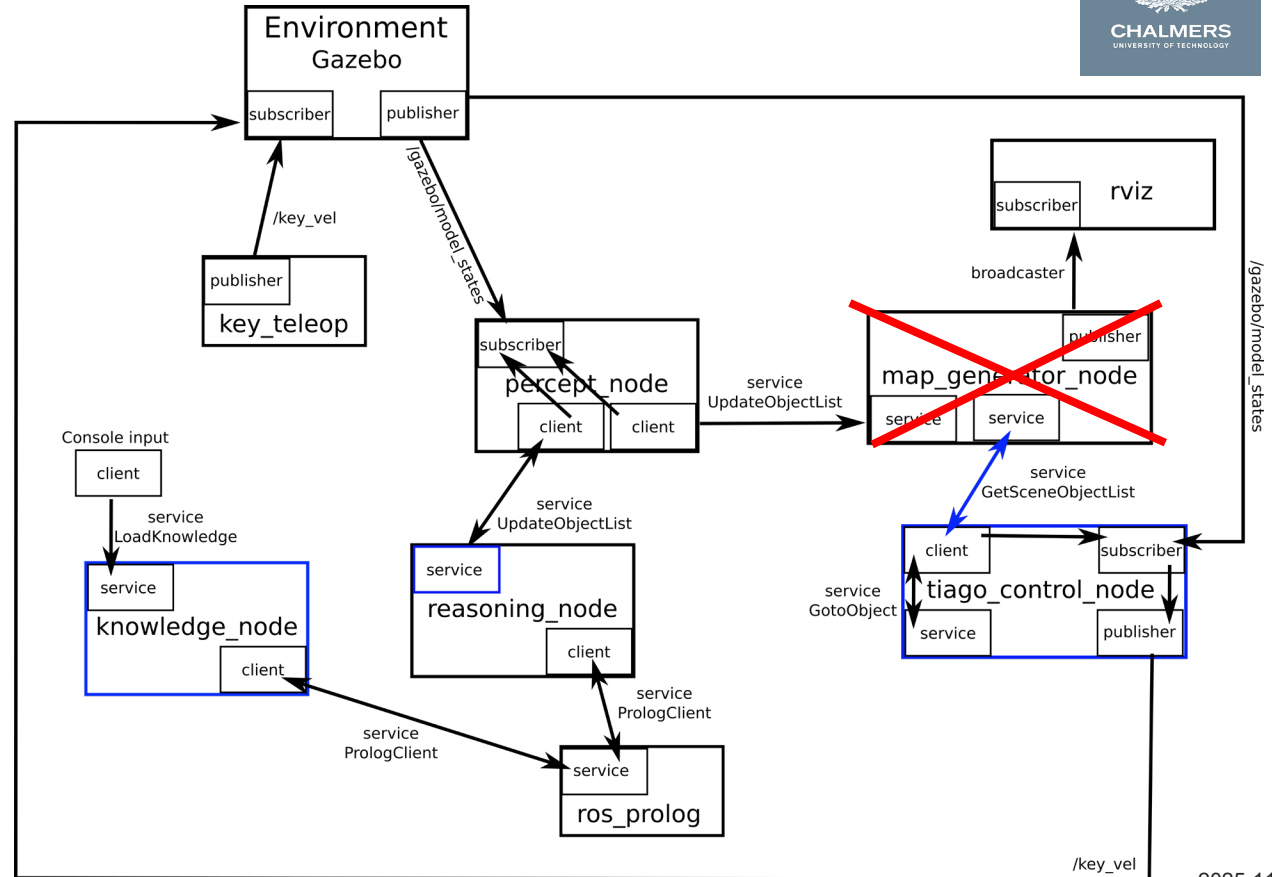
2) You need to include a new node “learning_node” that connects the nodes:

- percept_node
- reasoning_node
- tiago_control_node



Final Project

This means that the “tiago_control_node” should communicate with either the “percept_node” or the “reasoning_node” or both through the new “learning_node”. You need to decide which node is more appropriate to communicate with the control node.



Final Project



The final project consists of three main tasks. You can gain the following points per task:

Tasks	Task Description	Points
FP.T01: Reasoning	1) Implement at least four new Prolog predicates to infer more knowledge about the environment or the task. 2) Improve the robot's knowledge by including new knowledge on demand (using assertions) 3) Implement at least three new Prolog predicates to make decisions.	12
FP. T02: Learning	1) Implement at least one learning approach to solve your project's problem/task(s), e.g., Decision trees, Bayesian, ANN, RL, or other? 2) Use at least one online tool in your project (Movelt, ANN, OpenAI, Scikit, ...) 3) Implement at least one evaluation performance to measure the validity of your system, e.g., Confusion Matrix.	10
FP. T03: Robotics	1) Implement a more complex Perception-Action loop for the robot, for example, include at least one manipulation skill (e.g., pointing towards objects or grasping). Note: grasping has more complexity and gives more points. 2) Implement a perception system that uses the robot's sensors, e.g., cameras, lasers, ... You can use existing libraries, e.g., Aruco, OpenCV, Yolo, ...	10

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Note: You will only receive points for tasks T01, T02, and T03 if they are fully completed.

- This means that all requirements listed in the task description (slide 6) must be included in your solution.
- **Partial points will not be awarded for a task.**
- For example, if you provide the correct solutions for all three subtasks in T01, you will receive the full 12 points (even if you do not complete T02 or T03). However, if you miss even one subtask of T01, you will receive zero points. The same applies to tasks T02 and T03.
- You may submit any number of complete tasks. For instance, you can provide your solutions for T02 without submitting solutions for T01 or T03. In that case, only your solutions for T02 will be evaluated.

Final Project – Schedule



You should start working on the final project on Wed Dec 09. Please register before Dec 15 at the following link if you intend to deliver the FP. You **should choose one date to deliver your FP** solutions.

<https://docs.google.com/spreadsheets/d/1Zb9AC5EBpth5z30WWEktyjxhhc2kFsepbrZPffJ2qtE/edit?usp=sharing>

Study week	Monday	Tue	Wednesday	Thurs	Friday
6 (Dec 8-12)	10 am Q&A A4		10 am Q&A FP 1:15 pm Q&A FP		
7 (Dec 15-19)	10 am Q&A FP Deadline to register for FP		10 am Q&A FP		Deadline→ Date 1 to deliver FP
8 -10 (Dec 22–Jan 11) → Discussion Forum					
11 (Jan 13-17)					Deadline→ Date 2 to deliver FP

Assignment 04



You choose the deadline for submitting your solutions for the final project, as follows:

- 1) Dec 19 at 11:59 pm
- 2) Jan 16 at 11:59 pm

Please upload your original material before the deadline

- **Additionally, please upload a Video to show the simulation of your solutions**
- Upload your whole package in a zip file (named “final_project_teamNr.zip”) or provide the link to your git repo as a comment on Canvas. Please make sure to include a README file that explains how to run your code.
- Submit your group’s final evaluation report on Canvas.



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